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FROM: Lead BI Analyst

SUBJECT: Revenue Intelligence Dashboard Delivery & Analysis

Page 1: Architecture, Semantic Modeling & Dashboard Rationale

1. Data Strategy & System Architecture

- **Star Schema vs. Flat Joins:** Joining novatech_crm_deals.csv (499 rows), novatech_marketing_campaigns.csv (2,240 rows), and novatech_support_tickets.csv (3,000 rows) directly on account_id causes an M:N Cartesian fan-out of 65,448 rows, repeating every lead, deal, and ticket dozens of times. To maintain metric accuracy, each dataset is preserved as an independent fact table connected via a central Dim_Account table in a Star Schema.

- **Handling Orphan Accounts:** CRM contains 85 unique accounts (ACCT-001 through ACCT-085), whereas Marketing and Support include 15 additional orphan accounts (ACCT-101 through ACCT-115), representing 150 marketing touches and 204 support tickets. Direct INNER JOIN queries drop this data; FULL OUTER logic in Dim_Account prevents data loss.

- **Surrogate Key Generation:** Glitches upstream generated duplicate keys (3 duplicate opportunity_ids in CRM across 6 rows and 4 duplicate ticket_ids in Support across 8 rows). Composite surrogate keys (account_id + transaction ID) ensure 100% relational integrity.

- **Imputation & Cleansing:** The 24 missing values in Marketing annual_income (1.1%) were median-imputed by segment. The 59 open support tickets with null resolution dates were assigned a status flag of "Open".

2. Dashboard Design Rationale

- **View 1: Marketing Funnel:** Visualizes lead progression through five stages (Prospect→ Lead → Qualified Lead → Opportunity → Closed Won), spend vs. attributed revenue, and campaign-level ROI.

- **View 2: Sales Pipeline:** Tracks regional pipeline concentration, average deal value across company size tiers, win rates (64%), and stage outcomes.

- **View 3: Customer Health:** Surfaces operational friction by monitoring product areas, resolution times, sentiment, and an *At-Risk Accounts* matrix ranking high-ARR clients with elevated ticket volumes.

- **Interactive Navigation:** Cross-sheet actions carry account_id dynamically; clicking an account in Sales Pipeline immediately navigates to Customer Health filtered to that client's support backlog.

3. Amazon Q Topic Configuration & NLP Accuracy

- **Metric Disambiguation:** Mapped *revenue*, *sales*, and *deal bookings* strictly to deal_value to prevent Q from confusing company sales with client size (annual_revenue_usd) or marketing attribution (revenue_attributed).
- **Pre-Computed Formulas:** Built explicit calculated fields into the Topic for Win_Rate, Days_to_Close, and Resolution_Time_Days so plain-English operational queries evaluate reliably.
- **Geographic Aliasing:** Mapped *country* and *market* to Mkt_Src_Cd (two-letter ISO codes) to resolve ad-hoc regional queries.

Page 2: Key Insights, Actions & AI Validation

4. Quantified Findings & Recommended Actions

Operational Domain	**Quantified Finding**	**Business Implication**	**Recommended Action**
Acquisition Channels	**Partner Referral** generated **63.0% of all Closed Won leads (237 of ~376)** and 493 Qualified Leads. **Organic Search** generated 558 early leads but almost zero Closed Won outcomes.	Referral partners represent NovaTech's primary conversion engine; organic search attracts top-of-funnel traffic that does not buy.	Reallocate top-of-funnel inbound budgets into co-marketing and commission tiers for high-performing referral partners.
Marketing Spend & ROI	Marketing spent **$353.51M** to generate **$25.05M** in attributed revenue (-$328.46M net loss). **NovaEdge Awareness** spent **$55.45M** to yield just **$0.21M** in revenue.	Marketing operates at an extreme deficit due to unmonetized awareness campaigns.	Immediately freeze spend on NovaEdge Awareness; require all campaigns to show direct pipeline attribution.
Sales Monetization	**Small tiers ($1.67K)** and **Enterprise tiers ($1.66K)** show nearly identical average deal values, while **Large tiers lag at $1.33K**.	Enterprise and Large accounts are heavily under-monetized relative to their size, indicating loose discounting.	Implement mandatory minimum seat commits and bundle NovaPulse Ultimate into enterprise sales cycles.
Regional Execution	**West ($42.67M)** and **Central ($38.70M)** generate **80.6% of pipeline value**, while	Revenue generation is geographically unbalanced, signaling execution	Conduct a sales management review in the East and reassign proven account executives

Operational Domain	Quantified Finding	Business Implication	Recommended Action
	the **East ($19.53M)** lags significantly.	gaps in the Eastern territory.	from Central to drive pipeline.
Product Reliability	**Notifications** generated **19.98K tickets**, followed by **Authentication (11.31K)**. **Enterprise and Plus tiers absorb 96.7% of high-priority tickets (356 of 368)**.	Core operational friction impacts our highest-paying enterprise accounts, elevating renewal churn risk.	Deploy a dedicated tier-1 engineering escalation pod for Enterprise accounts and refactor the notifications service.

5. AI Analysis Validation (Dashboard vs. AI Findings)

- **Agreements:** AI narrative summaries and the dashboard charts fully aligned on Partner Referral conversion dominance, NovaPulse driving over 77% of sales, and Notifications/Authentication driving the largest ticket volume.
- **Disagreement on Lead Uniqueness:** AI analyzing the denormalized join reported that *"100% of leads are duplicated with 0 unique leads"*. The dashboard's distinct count aggregation (countDistinct(lead_id)) correctly identified that all 2,240 leads in the source file are distinct; the apparent duplication was a Cartesian side effect of flat joins.
- **Disagreement on Resolution Duration Sign:** AI summaries read the metric axis literally, reporting that *"average resolution time is -1.97 days"*. The dashboard formula had simply reversed the date subtraction order (created − resolved); actual operational resolution duration is positive 1.97 days (~47 hours).
- **Disagreement on Campaign Viability:** AI flagged all campaigns as complete failures based solely on immediate attributed revenue ($25.05M). The cross-view model shows that partner touches directly fed enterprise accounts responsible for $100.9M in closed pipeline, demonstrating that multi-touch LTV must be evaluated before cutting channels entirely.